

Pseudo-label data augmentation and graph-structured generative model for aspect sentiment triplet extraction

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ABSTRACT

Generative models have recently become a research focus due to their strong performance on the Aspect Sentiment Triplet Extraction task. Nevertheless, the performance of traditional generative models is limited. They often overlook the enhancement of performance through data augmentation and fail to fully utilize graph neural network features to analyze dependency syntax information, resulting in modest results, with average F1 scores of 71.94% and 71.47% on datasets ASTE-Data-v2 and ASTE-Data-v1, respectively. To address these challenges, this paper proposes a pseudo-label data augmentation and graph-structured generative model (PAG). PAG uses automatically generated pseudo-labeled data to enrich training samples and incorporates syntactic dependency information via graph-structured embeddings. The model is able to better capture sentence structure and relational information without relying on extensive manual annotations. Extensive experimental results on two public ASTE datasets show that the PAG model outperforms existing methods. Specifically, on the ASTE-Data-v2 dataset, our model improves F1 scores by 0.23%, 1.06%, 0.68%, and 0.20% on the 14lap, 14res, 15res, and 16res sub-datasets, respectively, resulting in an average improvement of 0.55%. On the ASTE-Data-v1 dataset, it improves F1 scores by 0.69%, 0.63%, 0.77%, and 0.23% for the corresponding sub-datasets, with an average improvement of 0.58%. Despite these improvements, the model's performance may still be influenced by the quality of pseudo-labeled data and the accuracy of syntactic dependency features, suggesting there is room for further refinement.

1. Introduction

Sentiment analysis [1–4], as a critical component of natural language processing, primarily focuses on sentiment classification tasks, such as determining whether the overall sentiment expressed in a text is positive or negative. As research advances, aspect-based sentiment analysis has emerged, which not only involves identifying specific aspects within a text but also classifying the sentiment expressed for each aspect [5–10].

Subsequently, the Aspect Sentiment Triplet Extraction (ASTE) task was introduced to further refine the aspect-based sentiment analysis framework [11–14]. ASTE requires not only the identification of aspects and sentiment terms from the text but also the delineation of the relationships between these terms, thereby offering a more comprehensive and in-depth sentiment analysis. An example of the ASTE task is shown in Fig. 1. The ASTE task requires the complete and accurate extraction of all triplets from a sentence, where each triplet consists of an aspect

term, an opinion term, and the sentiment polarity corresponding to the opinion term.

Initially, researchers proposed the ASTE task and used a two-stage pipeline method to extract triplets in phases [15]. The first stage involves identifying aspect terms and opinion terms in the sentence. The second stage determines the sentiment polarity relationship between the identified terms. Although the method is effective, it may lead to error accumulation and overlook the contextual relationships between terms. Therefore, subsequent research gradually shifted towards end-to-end models to better capture the associations between terms.

In subsequent research, some researchers continued to use Bidirectional Long Short-Term Memory (Bi-LSTM) [16] to extract semantic information from sentences for triplet extraction. By capturing contextual information within the sentences, Bi-LSTM can offer more accurate predictions when identifying triplets [15,17,18].

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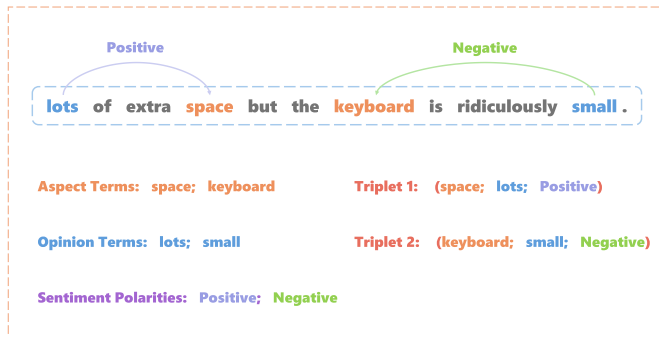


Fig. 1. An example of the ASTE task illustrates its objective: extracting structured triplets from a review sentence, each consisting of an aspect term, the corresponding opinion term, and the associated sentiment polarity. For instance, the sentence “lots of extra space but the keyboard is ridiculously small” contains two triplets: (space, lots, Positive) and (keyboard, small, Negative).

Following the rise of the pre-trained language model Bidirectional Encoder Representations from Transformers (BERT) [19], many researchers have used BERT to fine-tune for ASTE tasks, achieving excellent results [20–25]. BERT, with its bidirectional encoding capability, can deeply understand the contextual semantics of sentences, making it more precise in identifying aspect terms and opinion terms. Representative BERT-based methods have reported average F1 scores of 70.21% on ASTE-Data-v2 and 70.06% on ASTE-Data-v1, marking a clear improvement over earlier Bi-LSTM models.

Some researchers have noticed that Graph Neural Networks (GNNs) [26] can capture the dependency syntax information of sentences, effectively supplementing semantic information [27–31]. By representing sentence structures as graphs, GNNs can fully utilize the structured information in dependencies, helping the model better understand the grammatical and semantic relationships between words. These methods achieve average F1 scores of 70.15% on both ASTE-Data-v2 and ASTE-Data-v1, showing competitive performance with BERT-based approaches but without significant overall gains.

Finally, to extract triplets more conveniently and efficiently, some researchers have started using the generative model Bidirectional and Auto-Regressive Transformers (BART) [32] to generate triplets [6,33]. BART, with its encoder-decoder structure, can directly convert input sentences into the desired triplet output, effectively simplifying the task process. As generative models continue to evolve, the T5 [34] model, with its powerful generative capabilities, has gradually become mainstream in ASTE research [35–37]. The T5 model excels in triplet extraction tasks by handling various tasks through a unified text-to-text framework. Generative approaches have further advanced the state of the art, achieving average F1 scores of 71.94% on ASTE-Data-v2 and 71.47% on ASTE-Data-v1, though their performance remains only modestly higher than BERT-based and GNN-based methods, indicating that further improvement is still possible.

Although generative models have made significant breakthroughs in the ASTE task, most existing methods largely overlook the potential role of data augmentation in improving their performance. The data augmentation module can provide diverse training samples for the generative model, enhance its generalization ability, and improve the accuracy of triplet extraction. At the same time, the syntactic dependency information of a sentence is an important supplement to semantic information, as illustrated in Fig. 2. Previous generative models often lacked effective utilization of syntactic dependency information and did not integrate the syntactic information captured by GNNs with semantic information. The deep relationships between words may potentially not be fully understood by models when dealing with complex sentence structures.

To effectively address the above issues, we propose Pseudo-label data Augmentation and Graph-structure generative model (PAG), which

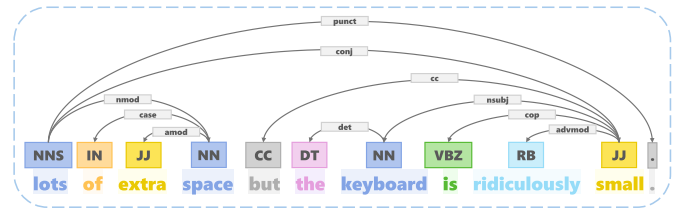


Fig. 2. Dependency structure of the sentence “lots of extra space but the keyboard is ridiculously small.” Each word is annotated with its part-of-speech tag, and arrows indicate dependency relations between words.

incorporates an innovative data augmentation strategy. Initially, we introduce two unlabeled datasets of the same type as the original dataset. Using an extraction model trained on the original dataset, we generate triples. Subsequently, we employ this extraction model to produce four candidate triple labels for both the original and unlabeled datasets. Based on the positions of the true labels in the candidate labels within the original dataset, a selection model is trained. A selection model is employed to identify suitable pseudo-label sequences from each sentence in the unlabeled dataset, thereby constructing a pseudo-labeled dataset. Finally, we combine the pseudo-label dataset with the original dataset to form an augmented dataset, enhancing the generalization capability of the generation model.

Additionally, we use Stanza¹ to parse the syntactic dependency information of sentences and employ an alignment algorithm to match it with the features of the generation model to better meet the model’s requirements. We employ a Graph Attention Network (GAT) [38] to capture syntactic dependency relations and extract graph-structured embeddings. These graph-structured embeddings are then embedded into the sentence’s encoded information, allowing the model to more deeply understand the relationship between syntactic dependency information and semantic information during the modeling process, thereby improving the overall performance and accuracy of the model.

The main contributions we have made are as follows:

- We propose a pseudo-label construction framework for unlabeled data, which employs extraction and selection models to generate and select high-quality pseudo labels.
- We integrate graph-structured embeddings into a generative ASTE model. By combining sentence encoding with graph-structured encoding and applying a token-level dependency alignment algorithm to adapt syntactic information to subword tokenization, our method achieves a dual representation of sentences.
- We conducted extensive experiments on two public ASTE datasets, and the results show that our model achieves state-of-the-art performance.

The structure of this paper comprises five main sections: Introduction, Related Work, Method, Experiments, and Conclusions. The Introduction presents the research background, motivation, and content. In the Related Work section, we summarize the various research methods currently employed in ASTE tasks. The Methodology section provides a detailed explanation of the proposed model’s framework and key components, offering an in-depth analysis of the model. The Experiments section validates the excellent performance of our method across various datasets through extensive experiments and provides a detailed analysis of the results. Finally, the Conclusion summarizes the core content of the research and offers insights into potential future research directions.

¹ <https://stanfordnlp.github.io/stanza/>.

2. Related work

Aspect-Based Sentiment Analysis (ABSA) has recently become an important direction in sentiment analysis, as it provides fine-grained insights at the aspect level and has attracted wide attention from the sentiment analysis research community [39–42]. Building on ABSA research, ASTE has become a significant research direction in the field of sentiment analysis. In this section, we will focus on introducing existing related works. Current ASTE methods can be mainly categorized into three types: BERT-based methods, graph neural network-based methods, and generative-based methods.

2.1. BERT-based methods

In the ASTE task, BERT has gained considerable attention due to its powerful language modeling and contextual understanding capabilities, attracting many researchers to use it for addressing ASTE challenges. Wu et al. [20] employed a unified grid tagging task and designed an effective inference strategy that leverages the mutual indication between different opinion factors. To jointly extract triplets, Xu et al. [21] employed a novel position-aware tagging scheme, using a sequence tagging approach within the joint model to extract these triplets. To reduce the high computational cost caused by span enumeration, Xu et al. [22] introduced a dual-channel span pruning strategy, integrating supervision from Aspect Term Extraction and Opinion Term Extraction tasks to enhance computational efficiency. Mukerjee et al. [23] employed a Pointer Network-based decoding framework, generating a complete opinion triplet at each time step. The decoder captured the interactions between targets and opinions by considering their entire detected spans. Zhang et al. [24] addressed ASTE through a novel relation span formulation where words and their semantic relations are jointly modeled, achieving comprehensive learning of aspect-opinion-sentiment triples through this unified representation. Liu et al. [43] delved into the deep and mutual interactive features between subtasks through two interaction modules and developed a General Word-Pair Relationship Learning component to refine the unreliable outputs of the first-level module. Zhao et al. [25] proposed the D2E2S model, which employs a dual-channel encoder mechanism. The BERT and enhanced LSTM channels are used to capture semantic and syntactic information, respectively. A heterogeneous feature interaction module was utilized to capture the complex interactions between dependency syntax and attention semantics. To distinguish the learning objectives of the two stages of the span-based method, Xu et al. [44] introduced part-of-speech information to assist span category prediction and dependency distance information to assist sentiment polarity category prediction. Lin et al. [45] proposed a span-level contextual interaction network by modeling bi-directional contextual interactions between aspect and opinion terms. Wang et al. [46] proposed an innovative approach called Decoupled Task Specificity, which involves constructing specialized term expert tables and sentiment expert tables, and designing task-specific decoders based on different region detection strategies.

2.2. Graph neural network methods

In the ASTE task, capturing complex semantic relationships and interdependencies is crucial for accurately extracting aspect terms, associated opinion terms, and sentiment polarities. Graph neural networks, due to their advantages in modeling relationships between nodes and handling unstructured data, are becoming the preferred choice for researchers. Chen et al. [27] designed a graph representation format that integrates syntactic dependencies, semantic relatedness, and positional relationships between words in a sentence, and encoded it using GraphSAGE [47]. Chen et al. [28] defined ten types of relationships and used a bi-affine attention module to embed these relationships as an adjacency tensor. Simultaneously, the sentence was transformed into a multi-channel graph to learn relation-aware node representations. To address

the issue of error propagation in previous methods, Li et al. [31] simultaneously identified aspect terms, corresponding opinion words, and associated sentiments in the final step to prevent error propagation across different subtasks. Shi et al. [48] combined contextual features with syntactic dependencies learned from the dependency graph through an interactive attention mechanism, engaging in iterative interaction to accurately capture semantic relationships. Yang et al. [30] used an unsupervised perturbed masking technique to extract syntax trees from pre-trained language models and employed a novel fusion strategy to integrate multiple features to capture the implicit connections among sentiment elements. Li et al. [49] proposed a dual-channel span generation method to constrain the search space of span candidates, while leveraging the syntactic relations among aspect or opinion terms and their part-of-speech characteristics to generate span candidates. To leverage contextual semantics and latent dependencies to enrich span representations, Jin et al. [50] designed a latent graph convolutional network and simultaneously detected effective aspect and opinion terms from the output of span enumeration. Yang et al. [51] used extraction modules equipped with two feature graph convolutional networks and original graph convolutional networks to enhance information exchange between nodes. Xu et al. [52] proposed a multi-level enhanced graph convolutional network that integrates sentiment and positional information with syntactic dependencies to improve triplet extraction accuracy.

2.3. Generative methods

In the ASTE task, generative approaches are gradually becoming an effective solution. Researchers have found that generative models do not rely on predefined labels and rules, and perform better when handling implicit information and complex sentence structures. Yan et al. [33] redefined each subtask as a sequence combining pointer indexes and sentiment class indexes, enabling all ABSA subtasks to be handled in a unified manner. Unlike traditional discriminative methods that require specific classification networks, Zhang et al. [6] introduced two modeling paradigms to transform each ABSA task into a text generation problem. To avoid incorrect triplet pairing, Zhou et al. [36] encoded aspects and opinions into two bidirectional templates and introduced a marker-oriented sequence labeling module to enhance the generative model's ability to handle complex structures. Fei et al. [53] modeled the ASTE task as an unordered triplet set prediction problem and utilized a non-autoregressive decoding paradigm with a pointer network. Deng et al. [35] utilized a generative model trained in both text-to-label and label-to-text directions, transforming each task into a unified format to learn domain-agnostic features, while generating natural sentences from noisy labels for data augmentation. Xu et al. [37] utilized a graph attention network to explicitly define the dependencies between words, while employing contrastive learning to capture the internal connections between sentiment types and term lengths.

3. Method

3.1. Problem description

In the generative aspect sentiment triplet extraction (ASTE) task, for a given sentence $S_i = \{w_1, w_2, \dots, w_n\}$ containing n ($n \geq 1$) words w , the goal of the task is to directly generate all the triplets contained in the sentence $T = \{A, O, S\}$. Each triplet T consists of an aspect term A , a corresponding opinion term O , and a sentiment polarity S for the opinion term, where the sentiment polarity S can be *positive*, *neutral* or *negative*.

3.2. Model overview

The overall architecture of the PAG model is shown in Fig. 3 and is designed to directly address the ASTE task defined above. It consists of four main parts: pseudo-label data construction, dependency matrix

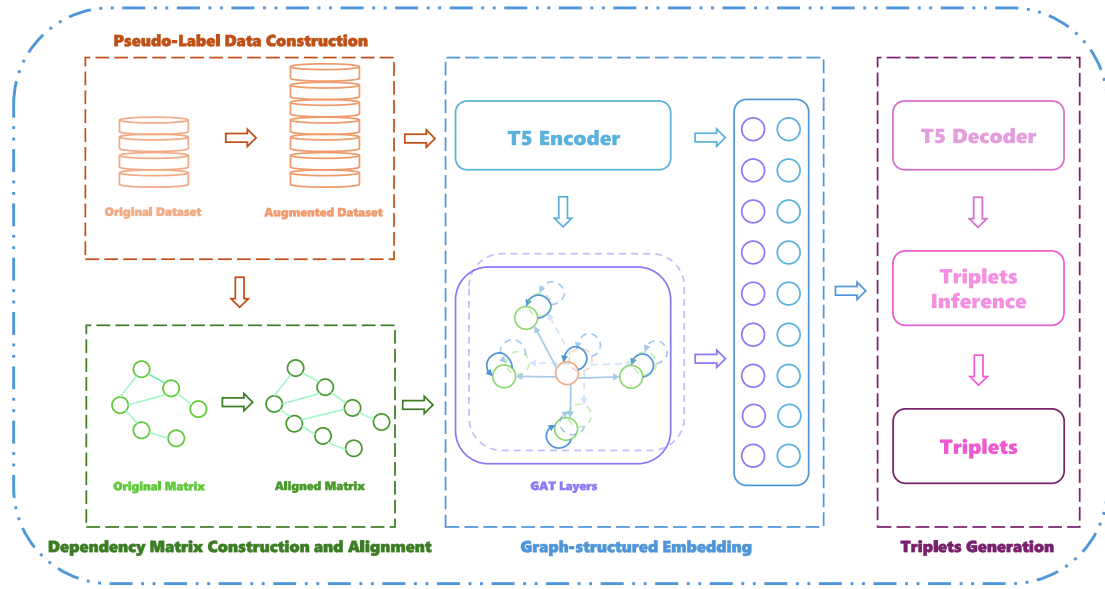


Fig. 3. The overall architecture of PAG, consists of four parts: pseudo-label data construction, dependency matrix construction and alignment, graph-structured embedding and triplet generation.

construction and alignment, graph-structured embedding, and triplet generation. The pseudo-label data construction module expands the training dataset by introducing high-quality pseudo-labeled examples, improving the model's ability to generate accurate triplets T . The dependency matrix construction and alignment component encodes syntactic dependencies from the sentence, allowing the model to capture the relationships between aspect and opinion terms. The graph-structured embedding module further integrates relational information from dependency structures into the T5 hidden representations, providing a richer sentence representation to support precise triplet extraction. Finally, the triplet generation component decodes this representation to generate the aspect, opinion, and sentiment polarity for each instance, thereby completing the mapping from sentence S_i to all triplets T required by the task.

3.3. Pseudo-label data construction

The core purpose of pseudo-label data construction is to expand the original dataset to enhance the generative capabilities of the model, as illustrated in Fig. 4. By innovatively utilizing the unlabeled dataset to generate pseudo-labels and training extraction and selection models, we achieve efficient generation of the pseudo-label dataset. The proposed approach enriches the diversity of training data and significantly enhances the model's generalization ability, ultimately resulting in improved performance in real-world applications.

3.3.1. Extraction model training

During the pseudo-label data construction process, the first step is training the aspect sentiment triplet extraction model. We utilize the T5

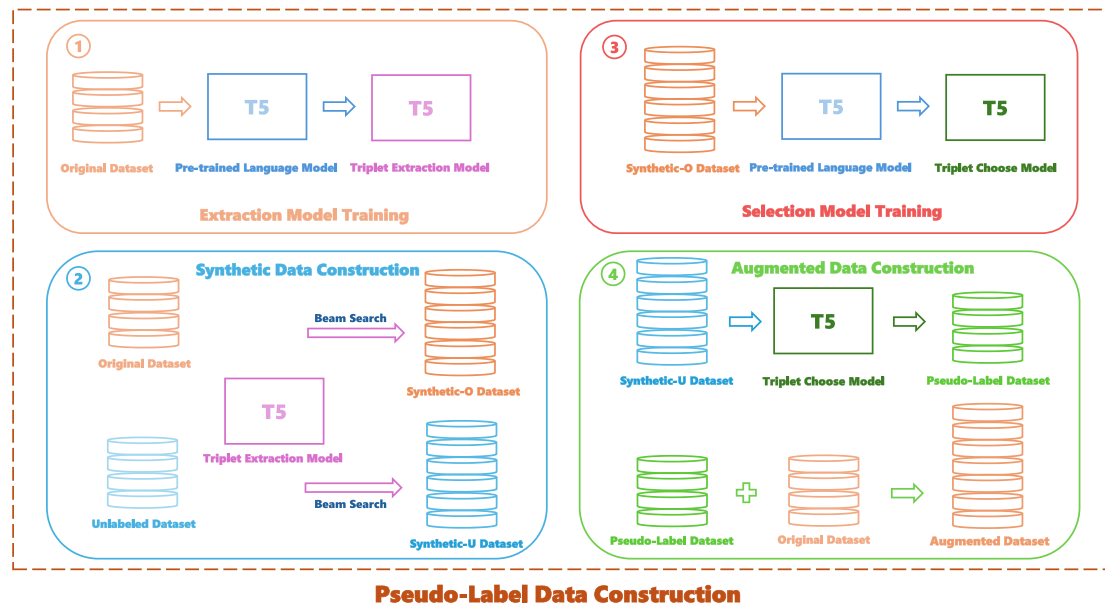


Fig. 4. Diagrammatic representation of pseudo-label data construction. The figure is divided into four steps: extraction model training, synthetic data construction, selection model training and augmented data construction.

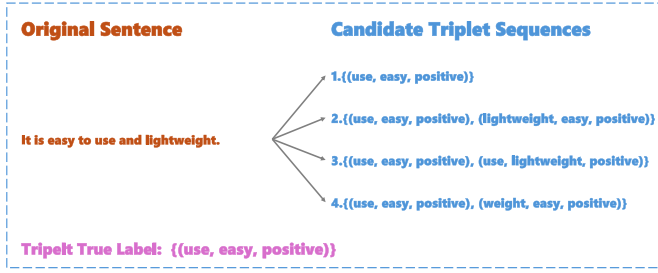


Fig. 5. Example of synthetic data. Each piece of synthetic data consists of one original sentence and a sequence of four candidate triplets.

pre-trained language model to train the triplet extraction model using the original dataset, aiming to optimize the model for precisely identifying aspect terms, opinion terms, and sentiment polarity triplets in sentences.

The cross-entropy loss for triplet extraction guides the model to generate the correct aspect term A , opinion term O , and sentiment polarity S for each sentence. During training, for a given input sentence S_i , the model predicts a triplet set $\hat{T}_i$ and compares it with the ground-truth triplet set T_i :

$$\mathcal{L}_{\text{Extraction}} = - \sum_{i=1}^N \sum_{t \in T_i} \log P(t|S_i) \quad (1)$$

where, i indexes the training samples, N represents the number of training samples, and $P(t|S_i)$ is the probability of the model generating the correct triplet t .

3.3.2. Synthetic data construction

At this stage, to maintain consistency with the coverage of the original dataset, we opt to use the restaurant dataset Yelp² and laptop dataset Laptop [54] as the unlabeled datasets. We exclusively use the original sentences from these datasets to generate pseudo-labels, ensuring that the generated labels are more closely aligned with the content of the original data. To effectively construct pseudo-labels for unlabeled datasets, we employ beam search decoding during the decoding phase to generate candidate triplet sequences, thereby fully leveraging the labeled information in the original dataset. The proposed approach enhances the model's generalization ability in unlabeled environments while substantially improving its robustness. Specifically, after initial training, we utilize a triplet extraction model to generate a synthetic dataset. By setting the beam width of the beam search decoding to 4, we generate four triplet candidate labels and decode them to produce triplet pseudo-label sequences for both the original dataset and the unlabeled dataset. These sequences are used to construct the original synthetic dataset Synthetic-O and the unlabeled synthetic dataset Synthetic-U. High accuracy and diversity in the generated triplet sequences are ensured by such a configuration. As shown in Fig. 5, each synthetic dataset entry not only includes the original sentence but also its corresponding four candidate triplet sequences.

3.3.3. Selection model training

To enhance the model's ability to select the most appropriate pseudo-label from four candidate triplet labels, we employ the T5 pre-trained model for training the selection model. The specific steps are as follows: Firstly, after obtaining the synthetic dataset Synthetic-O, we use the true label sequences from the original dataset to determine whether there is a matching label sequence among the four candidate triplet labels of Synthetic-O and identify its index position ($\{0, 1, 2, 3\}$). If no

matching label is found, the index is set to 4. Secondly, during training, cross-entropy loss is used to guide model optimization, ensuring accurate pseudo-label selection. In this manner, the selection model effectively leverages the language understanding and generation capabilities of T5, thereby improving the accuracy and efficiency of label selection.

The selection loss ensures that the model identifies high-quality pseudo-labeled data from multiple candidates, incorporating only reliable instances into the pseudo-label construction process. Assuming y is the index position of the true label (ranging from 0 to 4) and $\hat{y}_i$ is the predicted probability distribution, the loss is calculated as:

$$\mathcal{L}_{\text{Selection}} = - \sum_{i=0}^4 y_i \log(\hat{y}_i) \quad (2)$$

where, i indexes the label positions, y_i is the indicator for the true label (where the true label position is 1 and the rest are 0), and $\hat{y}_i$ is the probability distribution predicted by the model.

3.3.4. Augmented data construction

After completing the training of the selection model, the synthetic dataset Synthetic-U is subsequently predicted. The index produced by the model is utilized to identify the most suitable option among the four candidate triplets, thereby facilitating the efficient selection of the optimal pseudo-label. Pseudo-label sequences that yield empty predictions are then discarded, leading to a more precise and reliable pseudo-label dataset.

To better match the size of the original dataset in constructing the enhanced dataset and to avoid the interference of pseudo-labels with training effectiveness, we randomly select 1000 pseudo-labeled data points and integrate them with the original dataset. The proposed approach preserves the model's learning capability on the original data while simultaneously enhancing its generalization ability through moderate adaptation, with the addition of pseudo-labeled data contributing to improved overall model performance and accuracy.

3.4. Dependency matrix construction and alignment

As illustrated in Fig. 6, the construction of the dependency matrix begins with obtaining the syntactic dependency matrix of the sentence, followed by applying an alignment algorithm to produce the aligned matrix.

3.4.1. Syntactic dependency matrix construction

In the augmented dataset, sentences contain rich syntactic dependency information, which provides additional support for semantic understanding, thereby improving the model's performance during training. To effectively capture the implicit structural information in sentences and supplement the semantic capabilities of the T5 model, we use the syntactic analysis tool Stanza to perform dependency parsing on each sentence, resulting in the syntactic dependency matrix A . In matrix A_{ij} , an element of 1 indicates a dependency relationship between two words, while an element of 0 indicates no such relationship. For diagonal elements, when $i = j$, it represents the self-dependency of a word. The syntactic dependency matrix of a sentence can be expressed by the following formula:

$$A_{ij} = \begin{cases} 1, & \text{if } i \text{ and } j \text{ have a dependency} \\ 0, & \text{if } i \text{ and } j \text{ have no dependency} \\ 1, & \text{if } i = j \end{cases} \quad (3)$$

where, $A_{ij} \in \mathbb{R}^{n \times n}$, n represents the length of the sentence, and $0 \leq i \leq n-1, 0 \leq j \leq n-1$.

3.4.2. T5 token alignment algorithm

We obtain the syntactic dependency matrix A_{ij} of the sentence, which represents relationships between words. Since the T5 pre-trained

² <https://business.yelp.com/data/resources/open-dataset/>.

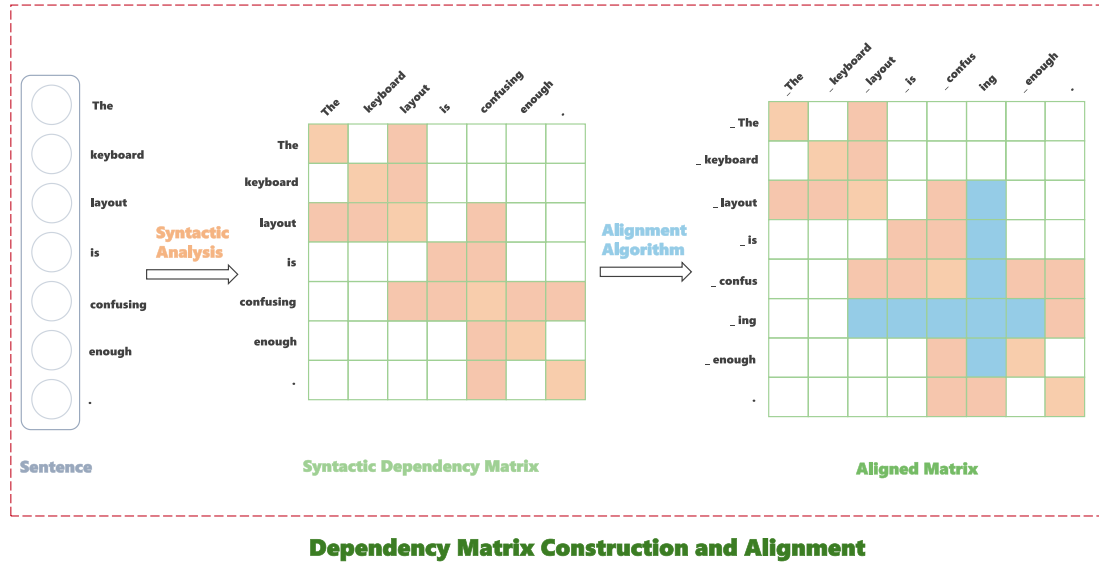


Fig. 6. Example of dependency matrix construction and alignment.

model operates on subword tokens rather than full words, we adapt a previously proposed BERT token alignment algorithm to the T5 tokenization scheme [55]. This alignment maps the original syntactic dependency matrix onto the tokenized sentence, ensuring that T5 can effectively leverage syntactic dependency information at the token level.

For example, consider a sentence “Tracking running activities is fun” with a dependency from “Tracking” to “running.” The T5 tokenizer may split “running” into “_run” and “_ing.” Our alignment algorithm ensures that both “_run” and “_ing” preserve the dependency with “Tracking,” allowing the model to capture the original syntactic relation despite subword tokenization.

As shown in Algorithm 1, the syntactic dependency matrix is represented using edge indices in PyTorch Geometric (PYG).³ The aligned matrix AL_{ij} is expressed as the token edge index. The alignment method preserves the syntactic dependency information of the sentence and ensures its effective utilization after tokenization. It aids the model in understanding dependencies between words at a fine-grained token level, thereby enhancing the capture of sentence semantics and context.

We note that for very long sentences, the construction and utilization of token-level edges may slightly affect computational efficiency, though the alignment still effectively maintains syntactic information for model training.

3.4.3. Graph-structured embedding

To more effectively capture syntactic dependency information within the alignment matrix, we utilize a Graph Attention Network (GAT) to extract deeper syntactic dependency features. Innovatively, these graph-structured embeddings are embedded into the hidden features of the T5 encoder. The detailed process of graph-structured embedding is illustrated in Fig. 7.

3.4.4. Feature coding

We utilize T5-base as the foundational framework for our model training, which consists of multiple encoders and decoders. To obtain the hidden feature representations $H_i^{encoder}$ from the encoder, we input only the sentence sequences S_i into the model to acquire the encoder’s output. The process can be expressed with the following formula:

$$H_i^{encoder} = T5 - Encoder(S_i) \quad (4)$$

Algorithm 1 Token Alignment with PyG Edge Index Construction.

Input: edge index EI , offset mapping O

Output: token edge index TEI

```

1: Initialize word positions  $WP \leftarrow []$ , current word  $CW \leftarrow []$ , previous
   end  $PE \leftarrow -1$ 
2: for each  $idx, (S, E) \in O$  do, where  $S$  represents start,  $E$  represents
   end
3:   if  $S \neq E$  then
4:     if  $S > PE$  and  $CW \neq []$  then
5:        $WP.add(CW)$ 
6:        $CW \leftarrow [idx]$ 
7:     else
8:        $CW.add(idx)$ 
9:     end if
10:     $PE \leftarrow E$ 
11:   end if
12: end for
13: if  $CW \neq []$  then
14:    $WD.add(CW)$ 
15: end if
16: Initialize  $TE \leftarrow [[]]$ 
17: for each  $(SN, EN) \in EI$  do, where  $SN$  represents start node,  $EN$ 
   represents end node
18:   for each  $ST \in WP[SN]$  do, where  $ST$  represents start token
19:     for each  $ET \in WP[EN]$  do, where  $ET$  represents end token
20:        $TEI[0].add(ST)$ 
21:        $TEI[1].add(ET)$ 
22:     end for
23:   end for
24: end for
25: return  $TEI$ 

```

3.4.5. Graph-structured extraction

Graph Neural Networks (GNNs) excel at capturing syntactic dependency information. They can comprehend complex structural information through the relationships between nodes and edges. Particularly, Graph Attention Networks (GATs) leverage their attention mechanism to dynamically adjust node weights. The proposed capability enables the model to more precisely identify and emphasize syntactic dependencies

³ <https://www.pyg.org>.

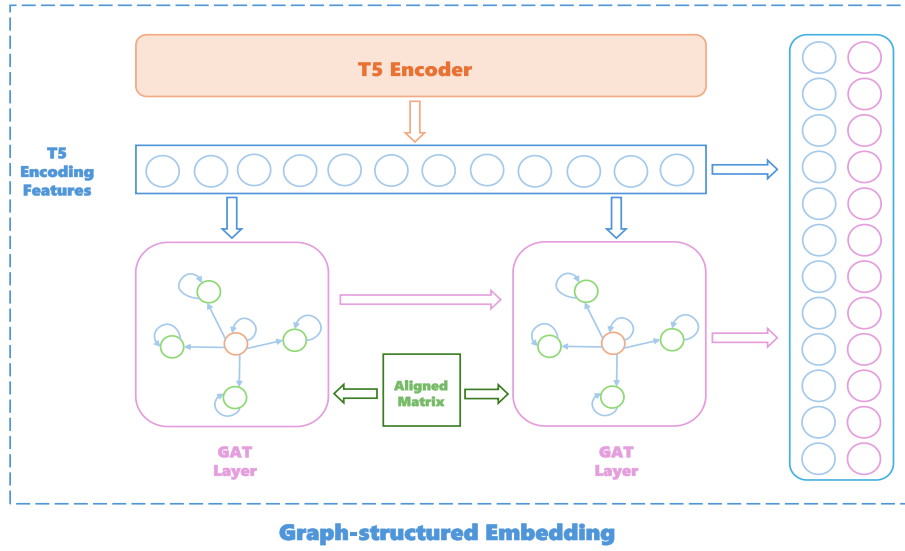


Fig. 7. Diagram of graph-structured embedding.

of key terms within sentences, consequently improving the overall effectiveness of feature extraction.

To fully utilize the syntactic dependency information in the alignment matrix AL , we employ a two-layer Graph Attention Network (GAT) to extract graph-structured embeddings from the encoded hidden features $H_i^{encoder}$. By integrating AL and $H_i^{encoder}$, GATs not only improve the precision of feature extraction but also enhance the model's understanding of complex syntactic relationships. Specifically, GAT initially extracts features T by combining AL and $H_i^{encoder}$. Then, by integrating AL with F , it extracts deeper graph-structured embeddings G . In this way, GAT not only improves the precision of feature extraction but also enhances the model's ability to comprehend complex syntactic relationships. The formula for obtaining graph-structured embeddings G is expressed as follows:

$$T = GAT_1(AL, H_i^{encoder}) \quad (5)$$

$$G = GAT_2(AL, T) \quad (6)$$

The formula for node feature updates in GAT is as follows:

$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W} \mathbf{H}_j \right) \quad (7)$$

where, α_{ij} is the attention weight between node i and node j , $\mathbf{W}$ is the trainable weight matrix, σ is the activation function, $\mathcal{N}(i)$ represents the set of neighboring nodes of node i obtained from the adjacency matrix AL , and $\mathbf{H}$ is a simplified representation of the encoded hidden features $H_i^{encoder}$.

The attention mechanism computes weights only between a node and its immediate neighbors in $\mathcal{N}(i)$, focusing on local syntactic dependencies. This design leverages the observation that direct neighbors capture most relevant information for a node, while considering all node pairs would increase computation and may introduce noise. The formula for calculating attention weights is as follows:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W} \mathbf{H}_i \parallel \mathbf{W} \mathbf{H}_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(\mathbf{a}^T [\mathbf{W} \mathbf{H}_i \parallel \mathbf{W} \mathbf{H}_k]))} \quad (8)$$

where, $\mathbf{a}^T$ is a trainable vector used to compute the attention scores, and $\parallel$ denotes the concatenation operation of vectors.

3.4.6. Feature embedding

After extracting graph-structured embeddings G , we integrate them into the hidden features of the T5 encoder $H_i^{encoder}$, enabling the model to simultaneously process multiple types of information. By facilitating the deep integration of syntactic and semantic information, this method strengthens the model's ability to interpret intricate linguistic phenomena. The formula for feature embedding is as follows:

$$F = H_i^{encoder} + G \quad (9)$$

3.4.7. Triplets generation

At this stage, we fine-tune the T5-base model using the complete augmented dataset to fully exploit the additional information and enhance model performance. The training objective follows the maximum likelihood estimation principle, where the model parameters are learned by minimizing the negative log-likelihood of the true sequences, thereby aligning the generated probability distribution with the ground truth distribution. Formulated as a conditional probabilistic model, the probability of generating a triplet sequence is defined over the entire vocabulary at each decoding step. For N training samples, each of sequence length T , the loss is computed as:

$$\mathcal{L} = - \sum_{i=1}^N \sum_{j=1}^T y_{ij} \log(p_{ij}) \quad (10)$$

where, N represents the number of training samples, T denotes the sequence length of each sample, y_{ij} is the true label of the i -th sample at the j -th position, and p_{ij} is the probability distribution predicted by the model for the i -th sample at the j -th position.

In the inference phase, we employ a beam search decoding strategy with the beam length set to 1 for efficient generation of triplets.

4. Experiments

4.1. Datasets

In this section, we use two publicly available datasets: D_1 (ASTE-Data-v1) and D_2 (ASTE-Data-v2), and assess the model's performance on these datasets. These datasets originate from the SemEval Task Challenge [56–58], a highly regarded international evaluation event in the field of sentiment analysis. Compared to D_1 , the D_2 dataset has been expanded and improved [22], incorporating more comprehensive annotation information and complex semantic structures, thus the

Table 1
Detailed statistics of the two datasets.

Dataset	Type	S	T	A	O
D_2	Train	3634	6205	5391	5550
	Dev	887	1511	1304	1354
	Test	1468	2536	2195	2244
D_1	Train	3655	5622	5379	5622
	Dev	909	1415	1363	1415
	Test	1478	2272	2194	2272

experiments primarily focused on D_2 . The statistical data of the two datasets are shown in Table 1, where S represents the number of sentences, T represents the number of triplets, A represents the number of aspect terms, and O represents the number of opinion terms.

4.2. Implementation details

In the construction of pseudo-label data and model training, we employed the T5 base model, which comprises 12 encoders and decoders, with a hidden state dimension of 768, and conducted training on the NVIDIA A30 GPU. In the GAT parameter settings, each hidden state dimension is set to 768, with the number of attention heads set to 1, effectively capturing syntactic features and embedding semantic information. During training, we used mixed precision training [59], optimized with the AdamW optimizer [60]. The training was configured for 20 epochs, with an initial learning rate of $2e-4$, applying a linear learning rate decay strategy, and setting the weight decay to 0.01 to enhance model performance and stability [61].

The experimental results are reported using Precision (P), Recall (R), and F1 score (F_1) based on the exact match of predicted triplets with the ground-truth triplets. Given the set of predicted triplets $\hat{T}$ and ground-truth triplets T , these metrics are calculated as:

$$P = \frac{|\hat{T} \cap T|}{|\hat{T}|}, \quad R = \frac{|\hat{T} \cap T|}{|T|}, \quad F_1 = \frac{2 \cdot P \cdot R}{P + R} \quad (11)$$

where, $|\cdot|$ denotes the number of triplets in the corresponding set.

To address the concern regarding statistical evaluation, we report all results as the mean F_1 , P , and R scores over five independent runs with different random seeds. In the ASTE task, the observed variability in these repeated experiments is minimal, so we chose to report only the mean values as a reliable measure of performance. This approach ensures stability while aligning with standard practices in this research direction.

4.3. Baselines

To validate the superiority of our proposed PAG model, this section will compare it against different types of baseline models. The selected baseline models are categorized into three major types: BERT-based models, graph neural network-based models, and generative-based models. The detailed descriptions of these baseline models are as follows:

1. BERT-based

- **HIM** [43] proposes an end-to-end Hierarchical Interaction Model that transforms the ASTE task into a unified boundary-words relation classification problem, achieving deep and mutual subtask interactions through multi-task learning and a Gated Interaction Network.
- **BDTF** [24] proposes a boundary-driven table-filling method, which represents triplets as relation regions in a two-dimensional table and addresses issues of relation inconsistency and boundary insensitivity for multi-word terms by optimizing table representation learning.
- **MiniConGTS** [62] designs a method that combines a minimalist tagging scheme with a novel contrastive learning strategy to

improve and utilize pre-trained representations through a more compact design and lower computational overhead.

- **IERET** [63] employs an implicit expression recognition enhanced table-filling method, which comprehensively captures word pair information by integrating overall implicit expression modeling in sentences and a bidirectional information aggregation module, and accurately delineates aspect-opinion boundaries during the decoding process.
- **DTS** [46] introduces an innovative approach called Decoupled Task Specificity, constructing term expert tables and sentiment expert tables for term extraction and sentiment classification, respectively, and designing task-specific decoders based on different region detection strategies.

2. Graph neural network-based

- **ESGAT** [30] designs an innovative edge-enhanced sentiment graph Attention Network model that transforms the impact matrix generated by the technique into new linguistic features and employs an effective fusion strategy to better capture the implicit connections among sentiment elements.
- **Dual-Span** [49] adopts a dual-channel span generation method that utilizes syntactic relations and part-of-speech characteristics of aspect/opinion terms to generate useful span candidates, thereby reducing span enumeration.
- **SpanGCN** [50] introduces a span-based dependency-enhanced graph convolutional network that enriches span representations through contextual semantics and latent dependencies, and handles various types of overlapping triplets.
- **MPGF-GCN** [51] proposes a multi-channel GCN network based on position-aware gating fusion, integrating multiple GCN modules and a position-aware gating fusion module to effectively extract aspect sentiment triplets.
- **MEGCN** [52] introduces a multiple-level enhanced graph convolutional network that enriches contextual understanding and positional analysis by integrating sentiment scores and polarity nodes, achieving a deeper representation of aspect sentiment triplets.

3. Generative-based

- **ASTE-BART** [33] defines each subtask target of ABSA as a sequence mixed with pointer indexes and sentiment class indexes, converting all subtasks into a unified generative formulation, and implements an end-to-end solution using BART.
- **PARAPHRASE** [64] jointly detects all sentiment elements in the form of quads and transforms them into a paraphrase generation process through a unique modeling paradigm.
- **SSI+SEL** [65] presents a unified text-to-structure generation framework, UIE, capable of universally modeling different information extraction tasks and capturing general information extraction abilities through a large-scale pre-trained model.
- **SLGM** [36] designs a sequence labeling enhanced generative model that encodes the dependency between aspects and opinions through bidirectional templates and introduces a marker-oriented sequence labeling module to handle complex structures.
- **NEDNF** [53] proposes an innovative encoder-decoder framework that effectively addresses overlapping structures and long-distance dependencies through a non-autoregressive decoding paradigm and a high-order aggregation mechanism.
- **GAC** [37] uses graph attention networks and contrastive learning to effectively identify aspects and opinions within complex structures, enhancing the ability to generate multi-word aspect and opinion terms.

4.4. Results

As shown in Tables 2 and 3, the experimental results of the PAG model on datasets D_2 and D_1 are presented. In the four sub-datasets

Table 2

The experimental results (%) on the D_2 dataset. The best results in P , R and F_1 are marked in bold, and “-” indicates not reported. The results with [‡] are retrieved from [36], while others are the results reported in the original paper.

Methods	14lap			14res			15res			16res		
	P	R	F_1	P	R	F_1	P	R	F_1	P	R	F_1
BERT												
HIM	65.99	56.05	60.59	76.99	70.46	73.57	69.65	63.23	66.26	73.11	71.05	72.06
BDTF	68.94	55.97	61.74	75.53	73.24	74.35	68.76	63.71	66.12	71.44	73.13	72.27
MiniConGTS	66.82	60.68	63.61	76.10	75.08	75.59	66.50	63.86	65.15	75.52	74.14	74.83
IERET	69.82	56.15	62.24	78.51	70.93	74.52	70.27	62.81	66.33	72.26	73.68	72.96
DTS	68.76	58.03	62.94	78.22	73.56	75.82	72.72	63.47	67.78	74.71	73.89	74.30
Graph												
ESGAT	61.11	58.96	60.01	72.43	74.05	73.23	65.53	63.50	64.50	67.36	68.81	68.08
Dual-Span	67.14	62.13	64.49	77.01	74.00	75.47	67.97	66.34	67.13	73.56	73.48	73.49
SpanGCN	72.37	52.96	61.16	77.99	68.69	74.04	64.33	66.90	65.05	72.50	70.30	71.38
MPGF-GCN	67.67	59.71	63.32	-	-	-	69.78	64.82	67.21	74.61	74.48	74.51
MEGCN	68.83	57.26	62.51	76.01	72.98	74.45	69.64	65.44	67.40	73.18	72.49	72.81
Generative												
ASTE-BART	61.41	56.19	58.69	65.52	64.99	65.25	59.14	59.38	59.26	66.60	68.68	67.62
PARAPHRASE [‡]	62.99	58.30	60.55	70.87	70.90	70.89	60.80	64.98	62.82	70.35	74.04	72.15
SSI + SEL [‡]	65.95	59.93	62.79	72.47	73.54	73.00	63.13	63.66	63.55	71.05	75.64	73.26
SLGM	70.54	60.74	65.27	78.84	72.70	75.64	69.75	66.85	68.27	75.86	75.76	75.80
GAC	71.31	62.25	66.47	79.11	73.14	76.01	70.12	68.00	69.04	76.62	75.88	76.25
PAG (Ours)	69.32	64.27	66.70	76.77	77.36	77.07	68.95	70.52	69.72	74.77	78.21	76.45

Table 3

The experimental results on the D_1 dataset. The best results in P , R and F_1 are marked in bold. The results are reported in the original paper.

Methods	14lap			14res			15res			16res		
	P	R	F_1	P	R	F_1	P	R	F_1	P	R	F_1
BDTF	68.30	55.10	60.99	76.71	74.01	75.33	66.95	65.05	65.97	73.43	73.64	73.51
MiniConGTS	67.45	61.01	64.07	75.87	76.12	76.00	66.84	64.08	65.43	69.38	74.40	71.80
DTS	68.34	57.26	62.31	77.74	75.44	76.58	68.82	66.21	67.49	74.03	73.67	73.85
ESGAT	61.19	58.16	59.64	72.88	71.78	72.33	65.90	63.89	64.88	69.72	71.31	70.51
MEGCN	69.64	57.22	62.78	77.36	74.57	75.94	69.47	65.45	67.32	73.72	75.44	74.54
Dual-Span	64.50	58.59	61.36	77.55	73.52	75.47	67.66	66.14	66.85	72.44	73.47	72.94
NEDNF	56.38	48.78	53.76	72.84	65.30	68.82	63.58	55.83	60.25	68.35	67.28	68.03
SLGM	67.91	58.57	63.08	77.99	73.26	75.55	69.28	65.84	67.52	74.90	75.49	75.20
GAC	69.14	59.82	64.14	79.25	73.82	76.44	70.40	67.54	68.94	76.95	75.80	76.37
PAG (Ours)	67.57	62.30	64.83	78.54	75.65	77.07	70.08	69.36	69.71	74.49	78.22	76.60

of D_2 , namely 14lap, 14res, 15res, and 16res, PAG achieves F_1 score improvements of 0.23%, 1.06%, 0.68%, and 0.20% respectively compared to the optimal model. Similarly, in the four sub-datasets of D_1 , PAG achieves F_1 score improvements of 0.69%, 0.63%, 0.77%, and 0.23%, respectively, compared with the best baseline model.

The experimental results from datasets D_2 and D_1 indicate that PAG achieves substantial improvements in average F_1 scores compared with BERT-based models, clearly demonstrating the advantages of generative modeling in ASTE tasks. Empirical evidence further suggests that generative models are better suited to capturing intricate semantic dependencies than traditional approaches. Moreover, compared with graph neural network-based models, PAG’s enhanced performance primarily stems from the integration of graph-structured embeddings, enabling

the model to capture complex relational structures more effectively. In addition, the incorporation of pseudo-labeled data further boosts triplet generation accuracy by expanding the training samples and introducing more diverse sentence structures. These findings highlight the significant complementary benefits of graph embeddings and pseudo-label augmentation in enhancing the overall performance of the generative model.

4.5. Ablation study

In this section, we conduct an ablation study on dataset D_2 to thoroughly analyze the impact of various modules and their key components on the overall performance of the PAG model. The experimental results of the ablation study are presented in Table 4.

Table 4

Results of ablation study on D_2 dataset. The best results are in boldface. “PDC” stands for pseudo-labeled data construction, “GAT” stands for graph attention network, “Align” refers to the alignment algorithm, and “GE” denotes the graph-structured embedding.

Methods	14lap			14res			15res			16res		
	P	R	F_1	P	R	F_1	P	R	F_1	P	R	F_1
w/o PDC	67.47	62.25	64.75	76.38	72.33	74.30	66.07	68.87	67.44	71.98	76.46	74.15
w/o GAT	68.15	62.62	65.26	75.48	74.45	74.96	68.39	68.04	68.21	74.57	75.88	75.22
w/o Align	68.59	63.54	65.97	75.77	76.66	76.21	68.99	69.07	69.03	74.91	77.24	76.05
w/o GE	67.83	64.09	65.91	76.58	75.45	76.01	69.50	69.07	69.29	75.68	75.68	75.68
PAG (Ours)	69.32	64.27	66.70	76.77	77.36	77.07	68.95	70.52	69.72	74.77	78.21	76.45

Firstly, from the data in the table, it is evident that the pseudo-labeled data construction has a significant impact on the performance of the PAG model. For generative models, the diversity of data is crucial to the generated results, hence the removal of the pseudo-labeled data construction leads to a noticeable drop in PAG's F1 score.

Secondly, when the graph attention network is removed, the performance of the PAG model also shows a significant decline. Graph attention networks demonstrate particular efficacy for this task by explicitly modeling sentence-level syntactic dependencies, enabling more reliable processing of pseudo-labeled datasets. Through this mechanism, the model can better parse sentence structures, thereby reducing interference from erroneous pseudo labels.

Subsequently, upon removing the alignment algorithm in the syntactic dependency matrix construction, we observe a slight performance decline in PAG. The alignment algorithm helps the graph attention network establish connections between tokens by matching the dimensions between T5 tokenization and syntactic dependency information, thereby enhancing the model's understanding of semantic information within T5's hidden features.

Lastly, the removal of feature embedding also results in a performance decline for the PAG model. The findings demonstrate that both graph-structured embeddings (capturing syntactic information) and encoded hidden features (representing semantic information) are essential for optimal model performance. Only by combining syntactic and semantic information can the model achieve a more accurate understanding and analysis of sentence structures.

4.6. Model analysis and visualization

In this section, we explore the factors affecting model performance in depth. We begin by analyzing the key components of the PAG model, specifically examining the impact of pseudo-label dataset composition and the role of GAT in graph-structured extraction. Subsequently, we perform attention visualization on the weights within the GAT for a more in-depth analysis.

4.6.1. Impact of pseudo-label dataset composition

During the pseudo-label construction phase, we randomly select 1000 synthesized pseudo-label data points from the dataset with the aim of enhancing the performance of the PAG model. To thoroughly investigate the impact of pseudo-label data on model performance, we conduct multiple experiments by adding 0, 500, 1000, 1500, and 2000 pseudo-label data points to the D_2 dataset. The experimental results are shown in Fig. 8.

Firstly, the experiment indicates that without adding pseudo-label data, the generation performance of the PAG model is poor. The generation model's limited ability to comprehensively learn sentence semantics under constrained data volumes directly impacts its generation performance.

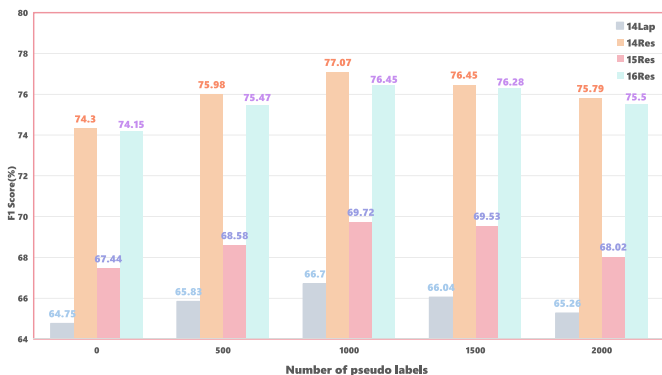


Fig. 8. F_1 score of different numbers of pseudo-labels in dataset D_2 .

Table 5

Detailed statistics of sentence counts for the two datasets. "Avg." stands for average.

Dataset	Sub-datasets	Sentences	Avg.Sentences
D_2	14lap	906	909
	14res	1266	
	15res	605	
	16res	857	
D_1	14lap	920	914
	14Res	1300	
	15res	593	
	16res	842	

As the number of pseudo-labels increases, the generation accuracy of the PAG model significantly improves, especially reaching optimal performance when 1000 pseudo-label data are added. Dataset expansion through moderate quantities of pseudo-labeled data proves sufficient to enhance semantic feature learning, as evidenced by the experimental results.

However, when the number of pseudo-labels is further increased, we observe a decline in the performance of the PAG model. Inherent label noise arises in the pseudo-labeled dataset as the extraction model must choose among four candidate labels, potentially introducing annotation errors. When there are too many pseudo-labels, these incorrect annotations can negatively affect the model's learning process.

Finally, we conduct a statistical analysis of the sentence counts in the sub-datasets of D_2 and D_1 , as shown in Table 5. The choice of 1000 pseudo-label data is because it is close to the average sentence count of D_2 and D_1 , allowing for an adequate expansion of the dataset size while ensuring balanced coverage of each sub-dataset. The optimal quantity of pseudo-labels achieves dual benefits: significantly boosting model performance while preventing the introduction of irrelevant noise and performance deterioration associated with excessive pseudo-labeling, ultimately optimizing the learning process.

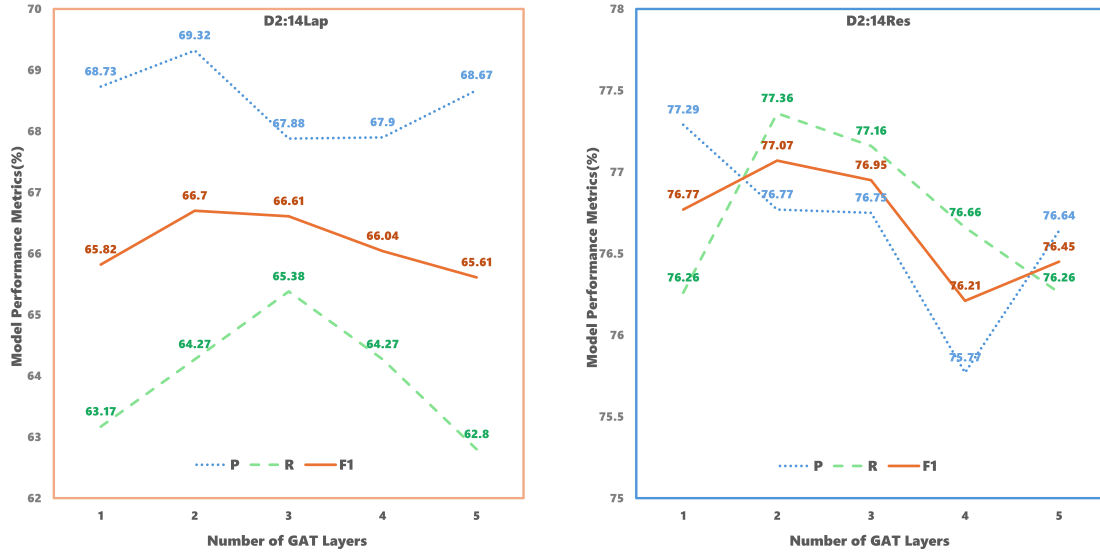
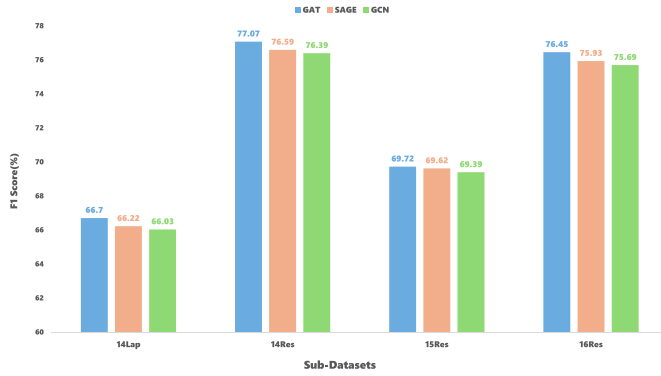
4.6.2. Impact of GAT layers

In the feature embedding module, we employ two layers of GAT for graph-structured extraction. To gain a deeper understanding of how the number of GAT layers affects model performance, we conduct comparative experiments on the D_2 dataset's sub-datasets, 14Lap and 14Res, as illustrated in Fig. 9. The results indicate that both subsets achieve optimal F_1 scores with two GAT layers, followed by a decline as the number of layers increases. The trend suggests that initially increasing the number of GAT layers can effectively enhance model performance, but beyond a certain critical number of layers, performance starts to drop. The performance degradation may stem from either excessive GAT layers causing model overfitting or over-concentrated information, both of which impair the model's generalization capability.

4.6.3. Impact of graph neural network types

In our PAG model, we employ a two-layer GAT to extract graph-structured embeddings. To evaluate the impact of different graph neural networks within the ASTE task, we also compare GAT with two commonly used alternatives: GCN and GraphSAGE, both configured with two layers. Fig. 10 presents the experimental results of different types of graph neural networks on the D_2 dataset.

The results indicate that GAT achieves the best performance in capturing complex relationships between nodes, followed by GraphSAGE and then GCN. The ASTE task requires identifying complex relationships between aspect and opinion terms. GAT can flexibly adjust the weights between nodes through its attention mechanism, allowing it to more precisely capture these complex relationships and generate triplets more accurately. SAGE mainly relies on aggregating information from neighboring nodes for feature extraction and lacks GAT's attention mechanism to dynamically adjust weights between nodes, making

Fig. 9. Model performance metrics on the D_2 dataset.Fig. 10. F_1 scores of different graph neural networks on the D_2 dataset.

it less flexible in capturing complex relationships and key information compared to GAT. GCN uses a simple weighted average mechanism to aggregate information from neighboring nodes, which lacks flexibility

and detailed processing capability, presenting challenges in recognizing complex node relationships.

4.6.4. Attention visualization in GAT

In GAT, attention weights can dynamically adjust, enabling the model to identify which words are most critical in recognizing the relationship between aspects and opinions. By enhancing syntactic structure analysis, the mechanism significantly improves extraction accuracy for both aspect terms and their corresponding opinion expressions. To visually demonstrate the role of attention weights, we selected the sentence “It is super easy to use.” and performed attention visualization on each layer of the two-layer GAT, as shown in Fig. 11.

In this attention visualization, we observe that both layers of the GAT give significant focus to the aspect term “use” and the opinion term “easy.” This is because the GAT can identify the critical role these words play in the sentence. In the second layer, the colors for “easy” and “use” deepen, indicating that the model is gradually concentrating attention on more important information. Our two-layer GAT design demonstrates selective attention capabilities—amplifying critical syntactic cues while attenuating non-essential elements, leading to accurate structural interpretation. Through the attention mechanism in GAT, the model can

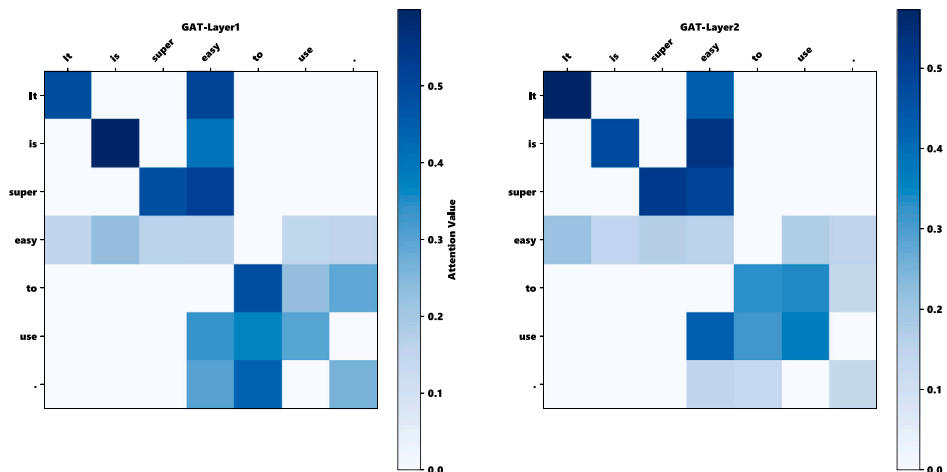


Fig. 11. Attention visualization of two-layer GAT.

Table 6

Case study of ASTE, where the correct result is marked as ✓, the wrong result is marked as ✗, and incorrect predictions indicate whether a triplet was missing or which component (aspect, opinion, sentiment) was incorrect.

Sentences	Ground truth	GAC	PAG (Ours)
1. And windows 7 works like a charm.	{{(windows 7, charm, POS)}}	{{(windows 7, charm, POS)✓}}	{{(windows 7, charm, POS)✓}}
2. Food is always fresh and hot ready to eat!	{{(Food, fresh, POS), (Food, hot, POS)}}	{{(Food, fresh, POS)✓, (Food, hot, POS)✓}}	{{(Food, fresh, POS)✓, (Food, hot, POS)✓}}
3. Even though the place is not beautiful, the food speaks for itself.	{{(place, not beautiful, NEG), (food, speaks for itself, POS)}}	{{(place, not beautiful, NEG)✓, (food, speaks for, POS)✗(opinion wrong)}}	{{(place, not beautiful, NEG)✓, (food, speaks for itself, POS)✓}}
4. Not only can the selection be innovative, but there's a nice balance of traditional sushi as well.	{{(selection, innovative, POS), (sushi, nice, POS), (sushi, traditional, POS)}}	{{(selection, innovative, POS)✓, (sushi, nice, POS)✓, ✗(missing triplet)}}	{{(selection, innovative, POS)✓, (sushi, nice, POS)✓, (sushi, traditional, POS)✓}}

more deeply analyze syntactic and semantic information, enhancing its performance in ASTE tasks.

4.7. Case study

In this section, we compare the proposed PAG model with the previous best model, GAC, on the same cases, as shown in Table 6. It can be seen that both PAG and GAC are generation-based models; therefore, when dealing with the first and second sentences which have relatively simple structures, both models can fully generate the correct sentiment triplets. In the third sentence, the PAG model can correctly identify the sentiment triplet “(food, speaks for itself, POS)”, whereas the GAC model fails to do so. This is because the PAG model uses pseudo-label data, making the training data more diverse and enabling the model to learn more similar sentence structures for more accurate results. In the fourth sentence, the PAG model can correctly identify the sentiment triplet “(sushi, traditional, POS)”, while the GAC model fails. PAG enhances its understanding of sentence syntax by embedding graph-structured embeddings into the T5 encoder’s hidden features, thereby improving its ability to analyze complex sentence structures.

5. Conclusions

This paper introduces an innovative generative model called PAG, which significantly enhances the accuracy of triplet generation by constructing pseudo-label data and embedding graph-structured embeddings into sentence encoding features. The introduction of pseudo-label data effectively expands the training data for the generative model, while a selection model automatically filters pseudo-label triplets, eliminating the need for labor-intensive manual annotation. During the syntactic dependency information construction phase, we employ an algorithm to dynamically align the syntactic dependency matrix with sentences tokenized by T5, ensuring effective correspondence between syntactic and semantic information. Additionally, we extract syntactic dependency information using GAT and innovatively embed graph features into the generative model for training, thereby fully supplementing the semantic information of the generative model with syntactic information from the graph-structured embeddings. In experiments, our model demonstrates state-of-the-art performance on two public datasets. Through ablation studies, we verify the effectiveness of each module and key component in PAG. We also analyze the impact of the pseudo-label dataset’s composition on the model results, proving the critical role of GAT in graph-structured extraction by using different layers of GAT and various types of GNN, and examining GAT’s ability to identify key information through attention visualization. In the future, we will explore the possibilities of data annotation and filtering within generative models, utilizing reinforcement learning for precise filtering of augmented datasets. Additionally, we will consider using prompt templates and large language models to generate pseudo-label datasets to further enhance the performance of triplet generation in generative models. Furthermore, we can introduce confidence calibration to more accurately assess the reliability of pseudo labels.

CRedit authorship contribution statement

Zheng Zhang: Writing – original draft, Methodology, Conceptualization. **Mingwei Tang:** Writing – review & editing, Funding acquisition, Data curation. **Jie Hu:** Validation, Formal analysis. **Li Wang:** Supervision, Resources. **Liansong Zong:** Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

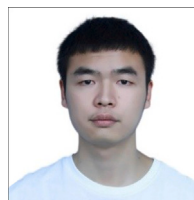
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